



Foodulator

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FOODULATOR



What is it?

- A tool that helps people figure out what they can cook based on the ingredients they already have at home.
- Saves money by reducing food waste
- Simplify planning for dinner, shopping list, meal prepping
- Lower the barrier of entry for new cooks



Previously.....

1. Dinner time rolls around.... Looks in the fridge to see what to cook
2. Has an idea of what you want to cook
3. Decide if you have enough ingredients to cook what you want
 - a. If you have ingredients, you search up recipe or start cooking
 - b. If you have to rethink what to cook and start the process again or go to store

Problem:

- Annoying.....
- Takes time and effort. => more likely to order takeout
- Have to plan grocery trips beforehand if want to reduce waste



Improved Flow

1. Dinner time rolls around.... Looks in the fridge to see what to cook
2. Has an idea of what you want to cook
 - a. Can check if you can cook what you want based on available ingredients
3. Notices something about to go stale
4. Utilizes Foodulator to get recommendations

Solution:

- Simplifies the decision making process
- promoting food waste reduction and saving money

Foodulator

Enter your ingredient:

Enter ingredient...

Tolerance:

Enter tolerance (e.g., 2)

New amount:

Enter new amount (e.g., 500)

Submit

Show Current Info

Update Amount

Responses:

Foodulator

Enter your ingredient:

butter

Tolerance:

Enter tolerance (e.g., 2)

New amount:

Enter new amount (e.g., 500)

Submit

Show Current Info

Update Amount

Responses:

Recipe ID: 1

Name: Cheeseburger Potato Soup

Directions: ["Wash potatoes; prick several times with a fork.", "Microwave them with a wet paper towel covering the potatoes on high for 6-8 minutes.", "The potatoes should be soft, ready to eat.", "Let them cool enough to handle.", "Cut in half lengthwise; scoop out pulp and reserve.", "Discard shells.", "Brown ground beef until done.", "Drain any grease from the meat.", "Set aside when done.", "Meat will be added later.", "Melt butter in a large kettle over low heat; add flour, stirring until smooth.", "Cook 1 minute, stirring constantly. Gradually add milk; cook over medium heat, stirring constantly, until thickened and bubbly.", "Stir in potato, ground beef, salt, pepper, 1 cup of cheese, 2 tablespoons of green onion and 1/2 cup of bacon.", "Cook until heated (do not boil).", "Stir in sour cream if desired; cook until heated (do not boil).", "Sprinkle with remaining cheese, bacon and green onions."]"]

Link: www.cookbooks.com/Recipe-Details.aspx?id=20115

NER: ["sour cream", "bacon", "pepper", "extra lean ground beef", "cheddar cheese", "green onion", "baking potatoes", "milk", "butter", "salt"]

Recipe ID: 2

Name: Rhubarb Coffee Cake

Directions: ["Cream sugar and butter.", "Add egg and beat well.", "To creamed butter, sugar and egg, add alternately buttermilk with mixture of flour, salt and soda.", "Mix well.", "Add rhubarb and vanilla.", "Pour into greased 9 x 13-inch pan and add Topping."]"]

Link: www.cookbooks.com/Recipe-Details.aspx?id=210288

NER: ["buttermilk", "egg", "sugar", "vanilla", "soda", "flour", "rhubarb", "butter", "salt"]

Recipe ID: 3

Foodulator

Enter your ingredient:
bacon

Tolerance:
Enter tolerance (e.g., 2)

New amount:
Enter new amount (e.g., 500)

Submit

Show Current Info

Update Amount

Responses:
Recipe ID: 1
Name: Cheeseburger Potato Soup
Directions: ["Wash potatoes; prick several times with a fork.", "Microwave them with a wet paper towel covering the potatoes on high for 6-8 minutes.", "The potatoes should be soft, ready to eat.", "Let them cool enough to handle.", "Cut in half lengthwise; scoop out pulp and reserve.", "Discard shells.", "Brown ground beef until done.", "Drain any grease from the meat.", "Set aside when done.", "Meat will be added later.", "Melt butter in a large kettle over low heat; add flour, stirring until smooth." "Cook 1 minute, stirring constantly. Gradually

Foodulator

Enter your ingredient:
bacon

Tolerance:
Enter tolerance (e.g., 2)

New amount:
Enter new amount (e.g., 500)

Submit

Show Current Info

Update Amount

Responses:
bacon is stored in the fridge with 10000.0g/ml.

Foodulator

Enter your ingredient:
bacon

Tolerance:
Enter tolerance (e.g., 2)

New amount:
0

Submit

Show Current Info

Update Amount

Responses:
Updated bacon to 0.0g/ml successfully.

Foodulator

Enter your ingredient:
bacon

Tolerance:
Enter tolerance (e.g., 2)

New amount:
Enter new amount (e.g., 500)

Submit

Show Current Info

Update Amount

Responses:
No matching recipes found with that ingredient and tolerance.

Foodulator

Enter your ingredient:

bacon

Tolerance:

1

New amount:

Enter new amount (e.g., 500)

Submit

Show Current Info

Update Amount

Responses:

Recipe ID: 1

Name: Cheeseburger Potato Soup

Directions: ["Wash potatoes; prick several times with a fork.", "Microwave them with a wet paper towel covering the potatoes on high for 6-8 minutes.", "The potatoes should be soft, ready to eat.", "Let them cool enough to handle.", "Cut in half lengthwise; scoop out pulp and reserve.", "Discard shells.", "Brown ground beef until done.", "Drain any grease from the meat.", "Set aside when done.", "Meat will be added later.", "Melt butter in a large kettle over low heat; add flour, stirring until smooth." "Cook 1 minute, stirring constantly. Gradually

Foodulator

Enter your ingredient:

cheddar cheese

Tolerance:

Enter tolerance (e.g., 2)

New amount:

0

Submit

Show Current Info

Update Amount

Responses:

Updated cheddar cheese to 0.0g/ml successfully.

Foodulator

Enter your ingredient:

bacon

Tolerance:

2

New amount:

Enter new amount (e.g., 500)

Submit

Show Current Info

Update Amount

Responses:

Recipe ID: 1

Name: Cheeseburger Potato Soup

Directions: ["Wash potatoes; prick several times with a fork.", "Microwave them with a wet paper towel covering the potatoes on high for 6-8 minutes.", "The potatoes should be soft, ready to eat.", "Let them cool enough to handle.", "Cut in half lengthwise; scoop out pulp and reserve.", "Discard shells.", "Brown ground beef until done.", "Drain any grease from the meat.", "Set aside when done.", "Meat will be added later.", "Melt butter in a large kettle over low heat; add flour, stirring until smooth " "Cook 1 minute, stirring constantly. Gradually



Future Vision

- Smart Grocery List Generator
 - Recommended grocery list based of ingredient amount and frequency of use
- Personalized Meal Recommendations
 - Recommends meals based off fitness and personal goals
- Mobile App Integration
 - Allows for barcode scanning to update storage
 - Adding recipe from social media

Far Future:

- Integration with a smart home system
 - Connected to health data and smart fridge

New Concepts: Partial matching



Regular queries vs Partial matching queries

- Does this match my criteria? vs Does this match my number of criteria?
- Instead of looking at direct matches, count the number of matches then compare

How?

- Logic: JOIN + GROUP BY + HAVING
 - Count how many required items match available ones, then filter by a threshold.
 - Threshold = tolerance
- Technique: Subquery-Based Filtering
 - Use one query to calculate some value, and another to compare to a different value



Outside Application

- This allows for flexibility, mimicking real-world thinking
- Applications:
 - Any recommendation systems
 - Job candidate recommendation
 - Online shopping recommendation
- Conceptual:
 - Spam/malware Detection
 - ML-based systems



